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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 16	
TITLE: User-Defined Exceptions	

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

1. **Create a Voting System that throws a custom exception if age is below 18**

Programme

```java
import java.util.Scanner;

// User-defined exception
class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

public class VotingSystem {

    static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("You are not eligible to vote. Age must be 18 or above.");
        }

        System.out.println("You are eligible to vote.");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter your age: ");
            int age = sc.nextInt();

            checkAge(age);
        }
        catch (InvalidAgeException e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output

```
PS C:\Users\RIKISHA> cd "C:\Users\RIKISHA\PROGRAMMING\Java\src"
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac VotingSystem.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java VotingSystem
Enter your age: 21
You are eligible to vote.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license

Programme

```java
import java.util.Scanner;

// User-defined exception
class InvalidDrivingAgeException extends Exception {
    public InvalidDrivingAgeException(String message) {
        super(message);
    }
}

public class DrivingLicense {

    static void checkEligibility(int age) throws InvalidDrivingAgeException {
        if (age < 18) {
            throw new InvalidDrivingAgeException(
                message: "You are not eligible for a driving license. Age must be 18 or above."
            );
        }

        System.out.println(x: "You are eligible for a driving license.");
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print(s: "Enter your age: ");
            int age = sc.nextInt();

            checkEligibility(age);
        }
        catch (InvalidDrivingAgeException e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output

```
Problems 9  Output  Debug Console  Terminal  Ports

PS C:\Users\RIKISHA> cd "C:\Users\RIKISHA\PROGRAMMING\Java\src"
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac DrivingLicense.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java DrivingLicense
Enter your age: 54
You are eligible for a driving license.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion

Thus, the programs successfully demonstrate the use of user-defined exceptions in Java by creating custom exception classes and handling them using try-catch. The programs validate age requirements for voting and driving licenses and display appropriate messages for invalid and valid inputs.